Aoxiang Fan

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Research Interests

I am dedicated to the study of **3D scene reconstruction and understanding** (related fields: *NeRF*, *Gaussian Splatting*, *Multi-view Stereo*, etc). The goal of my research is primarily to enable machines to better comprehend the world by modeling the environment intrinsically in 3D.

I am also interested and experienced in the matching problem ubiquitously existing in both vision and graphics (related fields: *image matching*, *graph matching*, *point cloud registration*, *shape matching*, etc).

Education

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland **2022-present**
Ph.D. in Computer Science, advised by Prof. ***Pascal Fua***

Electronic Information School, Wuhan University (WHU), China **2018-2021**
M.Eng. in Information and Communication Engineering, advised by Prof. ***Jiayi Ma*** *GPA:3.91/4.00*
Master Thesis (in Chinese): A Study of Robust Algorithms in Image Matching and Its Applications

Electronic Information School, Wuhan University (WHU), China **2014-2018**
B.Eng. in Electronic Information Science and Technology *GPA:3.50/4.00*

Selected Publications

1. **A View-consistent Sampling Method for Regularized Training of Neural Radiance Fields**
   Aoxiang Fan, Corentin Dumery, Nicolas Talabot, Pascal Fua
   Proc. International Conference on Computer Vision (ICCV), 2025

2. **Counting Stacked Objects**
   Corentin Dumery, Noa Ette, Aoxiang Fan, Ren Li, Jingyi Xu, Hieu Le, Pascal Fua
   Proc. International Conference on Computer Vision (ICCV), 2025

3. **Coherent Point Drift Revisited for Non-Rigid Shape Matching and Registration**
   Aoxiang Fan, Jiayi Ma, Xin Tian, Xiaoguang Mei, Wei Liu
   Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022

4. **Image matching from handcrafted to deep features: A survey**
   Jiayi Ma, Xingyu Jiang, Aoxiang Fan, Junjun Jiang, Junchi Yan
   International Journal of Computer Vision (IJCV), 2021

5. **Efficient Deterministic Search with Robust Loss Functions for Geometric Model Fitting**
   Aoxiang Fan, Jiayi Ma, Xingyu Jiang, Haibin Ling
   IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021

6. **Smoothness-Driven Consensus Based on Compact Representation for Robust Feature Matching**
   Aoxiang Fan, Xingyu Jiang, Yong Ma, Xiaoguang Mei, Jiayi Ma
   IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2021

7. **Geometric Estimation via Robust Subspace Recovery**
   Aoxiang Fan, Xingyu Jiang, Yang Wang, Junjun Jiang, Jiayi Ma
   Proc. European Conference on Computer Vision (ECCV), 2020

Preprints

1. **High-Fidelity and Generalizable Neural Surface Reconstruction with Sparse Feature Volumes**
   Aoxiang Fan, Corentin Dumery, Nicolas Talabot, Hieu Le, Pascal Fua
   arXiv preprint, 2025

Research Projects

A Foundation Model for Fine Geometry Recovery
*Supervised by Prof. **Pascal Fua*** *Perspective*

- *Background:* 3D reconstruction has been revolutionized by the advent of foundation models such as VGGT. However, fine-scale reconstruction at a single stage remains a challenge, due to both network architecture and hardware constraints.
- *Expected Contribution:* We aim to develop a novel method that performs second-stage refinement given rough geometry from foundation models. The method will focus on high-resolution and fine-scale feature learning, employing a sparse volume processing architecture to reduce memory restrictions.

Research Internship

TuSimple-Autonomous Trucking Technology, Beijing
*Supervised by Dr. **Ji Zhao** and Dr. **Naiyan Wang*** *November 2020-March 2021*

- Improved the localization accuracy of the autonomous vehicle by developing an outlier-resilient method for landmark-based 2D-image to 3D-point-cloud alignment.

Technical Strengths

- **Mathematics:** 3D Geometry, Linear Algebra, Numerical Optimization
- **Programming:** Python, C/C++, MATLAB, LaTeX
- **Developments:** Linux, Git, CMake, Docker, Kubernetes
- **Libraries:** PyTorch, OpenCV, Trimesh
- **Machine Learning:** Neural Rendering, 3D Gaussian Splatting, Generative AI
- **Language Skills:** Chinese (*Native*), English (*Proficient*, TOEFL 108/120), French (*Elementary*)

Awards

- Recipient of National Encouragement Scholarship of China 2017
- Second Prize Winner of the 17th China Post-Graduate Mathematical Contest in Modeling 2020
- Recipient of Outstanding Master's Thesis by the China Institute of Communications 2022
- Recipient of EDIC Fellowship, EPFL 2022

Additional Experiences

Conference Reviewer

Five-year experience reviewing for : CVPR, ICCV, ECCV, NeurIPS, ICLR.

Teaching Assistant

- CS-442 Computer Vision (*Head TA*) - Lead 8-10 assistants to create course material, teach and evaluate over 190 students. (2023, 2024)
- CS-401 Applied Data Analysis - Help supervise student projects, create course materials and grade exams. The ADA course is one of the largest at EPFL with over 700 students. (2023, 2024, 2025)

Student Supervision

- Boris Zhestiankin (M. Sc.) - *Learning Unsigned Distance Functions from Gaussian Splatting*; Now intern at Inria, Grenoble France.
- Yann Savioz (M. Sc.), - *Gaussian Spaltting Applied at City-scale with Lausanne Dataset*; TBD.